

COMP 5350/6350 Hands-On #2 Report

Student Name: Eristen Forsythe

1. Executive Summary

This report documents the forensic examination of the disk image case001_investigation_disk.dd from a fictional art theft case. The goal of the analysis was to reconstruct the suspects' plan and manually recover evidence files from both an NTFS partition and a FAT16 partition. By manually examining the MBR, partition boot sectors, directory structures (including the FAT16 root directory), and NTFS MFT records, I was able to identify and recover several key pieces of evidence. These included encoded location information, a password-protected archive containing the Certificate of Authenticity, and a second archive with hotel/room details and timing instructions. All recoveries were performed using dd and then verified with hexdump, file signature checks, and hash values.

Questions	Answers															
Q1) Identify the number of partitions on the disk image, and state the type of each partition.	The disk image has two partitions, one NTFS & FAT16. Partition 1: NTFS (MBR type 0x86), start sector 2048. Partition 2: FAT16 (MBR type 0x06), start sector 206848.															
Q2) For each partition, list every file you find, including file name and file size.	NTFS: - Ristorante-Italiano.png – 2,121,324 bytes (active) - registrar.zip – 92,570 bytes (active; recovered as registrar.zip) FAT16: - LOCATION.txt – 98 bytes (active) - COA.ZIP – 1,744,670 bytes (deleted; encrypted)															
Q3) For each file, report the starting byte offset and ending byte offset on the disk (where the file’s data is located). Use the recovery template Excel sheet as part of your submission.	<table><tr><th>File</th><th>start</th><th>end</th></tr><tr><td>Ristorante-Italiano.png -</td><td>55,574,528</td><td>57,695,851</td></tr><tr><td>registrar.zip -</td><td>72,351,744</td><td>72,444,313</td></tr><tr><td>LOCATION -</td><td>106,172,416</td><td>106,172,513</td></tr><tr><td>COA.ZIP -</td><td>106,174,464</td><td>107,919,133</td></tr></table>	File	start	end	Ristorante-Italiano.png -	55,574,528	57,695,851	registrar.zip -	72,351,744	72,444,313	LOCATION -	106,172,416	106,172,513	COA.ZIP -	106,174,464	107,919,133
File	start	end														
Ristorante-Italiano.png -	55,574,528	57,695,851														
registrar.zip -	72,351,744	72,444,313														
LOCATION -	106,172,416	106,172,513														
COA.ZIP -	106,174,464	107,919,133														
Q4) Manually recover all files from each disk image. Document the recovery with step-by-step instructions and annotated screenshots.	See Section 4.1.5 (NTFS) and Section 4.2.5 (FAT16) for complete manual recovery steps, commands, and screenshots.															
Q6) Analyze the recovered files in depth to determine what they reveal about the	LOCATION decodes to “Le Burgundy Paris” and GPS coordinates. registrar.zip contains images indicating hotel room 332 and a move time of 12:40 PM.															

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operation (objective, plan, key details).	COA.ZIP decrypts with password 75018 and contains a disguised PDF Certificate of Authenticity for the artwork “Sunset on Ice.”
Q7) Identify and explain any data hiding methods used on the laptop.	Base64-encoded text (LOCATION), Password-protected ZIP (COA.ZIP), Deleted-file hiding in FAT16 (COA.ZIP entry), Nested archive (registrar), Extension spoofing (sunset on ice.png is a PDF).
Q8) State the criminals’ ultimate objective, then build a timeline of their plan based on evidence.	Objective: obtain a photographic copy of the Certificate of Authenticity to sell the stolen artwork. Timeline: meeting location in Paris (ZIP 75018), registrar tracked to Le Burgundy hotel, move at 12:40 PM to room 332, CoA captured and stored in encrypted archive. See Sections 6–8 for the full narrative with supporting artifacts.

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2. Investigation Methodology (General Overview)

This investigation followed a manual, repeatable workflow and intentionally avoided automated forensic suites. The disk image was handled as read-only evidence and first validated with hash values. Partition identification was then completed by manually parsing the MBR partition table using `dd` and `hexdump`. For the FAT16 partition, I parsed the boot sector fields to calculate the locations of the FAT, Root Directory, and Data Area. I then examined the root directory with `hexdump` to identify file entries and determine their starting clusters. For the NTFS partition, I parsed the boot sector to locate the \$MFT, then extracted and interpreted specific MFT records to identify file names and non-resident data runs. Using the calculated sector offsets, I recovered files with `dd` and validated each recovery with `hexdump` signature checks, the file command, and cryptographic hashes. I also captured screenshots for every command and its full output, and recorded all sector, cluster, and byte offset calculations in `RecoverySheet.xlsx`.

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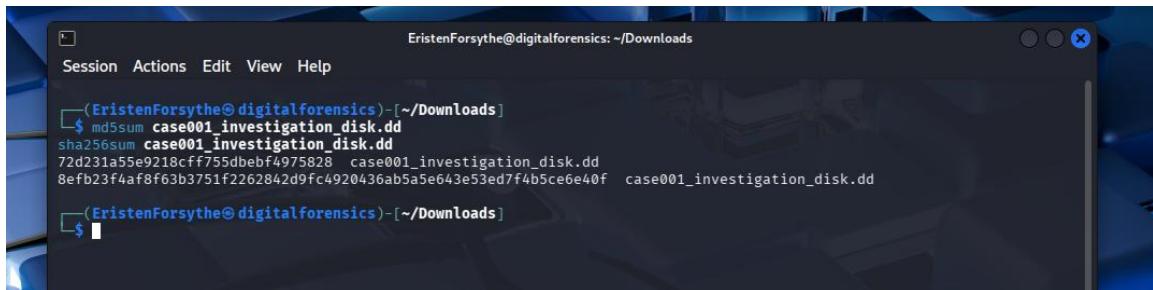
3. Disk Image Overview

3.1 Disk Image Properties

- Image format (DD): Raw DD (case001_investigation_disk.dd)
- Total disk size: 314,572,800 bytes (300 MiB)
- Sector size: 512 bytes
- Hash values (MD5/SHA-1/SHA-256)

MD5: 72d231a55e9218cff755dbebf4975828

SHA-256: 8efb23f4af8f63b3751f2262842d9fc4920436ab5a5e643e53ed7f4b5ce6e40f



```
ErystenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help
(EristenForsythe@ digitalforensics) - [~/Downloads]
$ md5sum case001_investigation_disk.dd
sha256sum case001_investigation_disk.dd
72d231a55e9218cff755dbebf4975828 case001_investigation_disk.dd
8efb23f4af8f63b3751f2262842d9fc4920436ab5a5e643e53ed7f4b5ce6e40f case001_investigation_disk.dd
(EristenForsythe@ digitalforensics) - [~/Downloads]
$
```

3.2 Partition Layout Summary

Partition	Type	Start sector/ Offset	End sector	Size (bytes)
1	NTFS (0x86)	2048 / 1048576 bytes	206847 / 105906175 bytes	104857600
2	FAT1 6 (0x06)	206848 / 105906176 bytes	452607 / 231735295 bytes	125829120

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```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

(EristenForsythe@digitalforensics)~/Downloads
$ fdisk -l case001_investigation_disk.dd
Disk case001_investigation_disk.dd: 300 MiB, 314572800 bytes, 614400 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xad5406c4

Device           Boot  Start    End  Sectors  Size Id Type
case001_investigation_disk.dd1      2048 206847   204800  100M 86 NTFS volume set
case001_investigation_disk.dd2    206848 452607  245760  120M  6 FAT16

(EristenForsythe@digitalforensics)~/Downloads
$
```

```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

(EristenForsythe@digitalforensics)~/Downloads
$ dd if=case001_investigation_disk.dd bs=512 count=1 | hexdump -C | head -n 40
dd if=case001_investigation_disk.dd bs=1 skip=446 count=64 | hexdump -C
00000000  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
000001b0  00 00 00 00 00 00 00 00 c4 06 54 ad 00 00 00 20 |.....T...|
000001c0  21 00 86 df 13 0c 00 08 00 00 20 03 00 00 df |!.....|
000001d0  14 0c 06 2c 10 1c 00 28 03 00 00 c0 03 00 00 00 |...,...(.....|
000001e0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
000001f0  00 00 00 00 00 00 00 00 00 00 00 00 55 aa |.....U.|
00000200
1+0 records in
1+0 records out
512 bytes copied, 0.00113342 s, 452 kB/s
64+0 records in
64+0 records out
00000000  00 20 21 00 86 df 13 0c 00 08 00 00 20 03 00 00 |. !..... ..|
64 bytes copied, 0.00183947 s, 34.8 kB/s
00000010  00 df 14 0c 06 2c 10 1c 00 28 03 00 00 c0 03 00 |.....,...(.....|
00000020  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
00000040

(EristenForsythe@digitalforensics)~/Downloads
$
```

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4. Partition-by-Partition Analysis

4.1 Partition 1 Analysis (NTFS)

4.1.1 Partition Description

- File system type: NTFS
- Starting and ending offsets: Sector 2048 (1048576 bytes) & Sector 206847 (105906175 bytes)
- Intended purpose (if inferable): Storage of operational communications and artifacts related to tracking the registrar and storing recovered evidence.

4.1.2 NTFS Structure Analysis

NTFS boot sector fields were extracted at the partition start (sector 2048). Bytes/sector and sectors/cluster were used to compute the cluster size. The \$MFT starting cluster (offset 0x30) was used to compute the \$MFT starting sector: $MFT_start_sector = NTFS_start_sector + (MFT_cluster_start \times sectors_per_cluster)$. MFT record size (1024 bytes) was used to extract individual records (2 sectors each). Screenshots inserted below.

Screenshot: NTFS boot sector at partition start (sector 2048).

```
ErystenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

(EristenForsythe@digitalforensics)-[~/Downloads]
$ dd if=case001 investigation disk.dd bs=512 skip=2048 count=1 | hexdump -C | head -n 80
1+0 records in
1+0 records out
512 bytes copied, 0.00140769 s, 364 kB/s
00000000 eb 52 90 4e 54 46 53 20 20 20 20 00 02 08 00 00 |.R.NTFS|.....|
00000010 00 00 00 00 00 00 f8 00 00 00 00 00 00 00 00 00 |.....|
00000020 00 00 00 00 00 80 00 00 00 ff 1f 03 00 00 00 00 00 |.....|
00000030 04 00 00 00 00 00 00 00 ff 31 00 00 00 00 00 00 00 |.....|
00000040 f6 00 00 00 01 00 00 00 88 81 a9 76 fc ba 44 0c |.....v..D.|
00000050 00 00 00 00 0e 1f be 71 7c ac 22 c0 74 0b 56 b4 |.....ql..t.V.|
00000060 0e bb 07 00 cd 10 5e eb f0 32 e4 cd 16 cd 19 eb |.....^..2.....|
00000070 fe 54 68 69 73 20 69 73 20 6e 6f 74 20 61 20 62 |.This is not a bl|
00000080 6f 6f 74 61 62 6c 65 20 64 69 73 6b 2e 20 50 6c |ootable disk. Pl|
00000090 65 61 73 65 20 69 6e 73 65 72 74 20 61 20 62 6f |ease insert a bol|
000000a0 6f 74 61 62 6c 65 20 66 6c 6f 70 70 79 20 61 6e |otable floppy an|
000000b0 64 0d 0a 70 72 65 73 73 20 61 6e 79 20 6b 65 79 |d..press any key|
000000c0 20 74 6f 20 74 72 79 20 61 67 61 69 6e 20 2e 2e | to try again ..|
000000d0 2e 20 0d 0a 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
000000e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
+
000001f0 00 00 00 00 00 00 00 00 00 00 00 00 55 aa |.....U.|
00000200
```

NTFS signature:

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4e 54 46 53 = NTFS = confirms NTFS filesystem

Bytes/Sector:

00 02 (little-endian) = 0x0200 = 512 bytes/sector

Sectors/Cluster:

08 = 8 sectors/cluster

Cluster size = $8 \times 512 = 4,096$ bytes

\$MFT starting cluster (LCN) @ offset 0x30:

04 00 00 00 00 00 00 00 (little-endian) = 0x0000000000000004 = 4

Derived:

Partition start sector = 2048

\$MFT start sector = $2048 + (4 \times 8) = 2080$

Region	Start Sector	End Sector	Size (sectors)	Start Offset (bytes)
NTFS Boot Sector (Partition Start)	2,048	2,048	1	1,048,576
Reserved / System Area (pre-\$MFT)	2,049	2,079	31	1,049,088
\$MFT (starts at LCN 4)	2,080	2,080	1	1,064,960
NTFS Data Area (general file data region)	2,081	206,847	204,767	1,065,472

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Bytes/sector = 512

Sectors/cluster = 8 (cluster size = 4,096 bytes)

\$MFT LCN = 4 = \$MFT start sector = $2048 + (4 * 8) = 2080$

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4.1.3 Root Directory Analysis

NTFS does not use a FAT-style fixed root directory. Instead, file and directory metadata are stored in the Master File Table (\$MFT). For this partition, the NTFS boot sector showed that the \$MFT begins at sector 2080, and each MFT record is 1024 bytes (2 sectors).

MFT start sector = 2080 (from NTFS boot sector)

MFT record size = 1024 bytes = 2 sectors

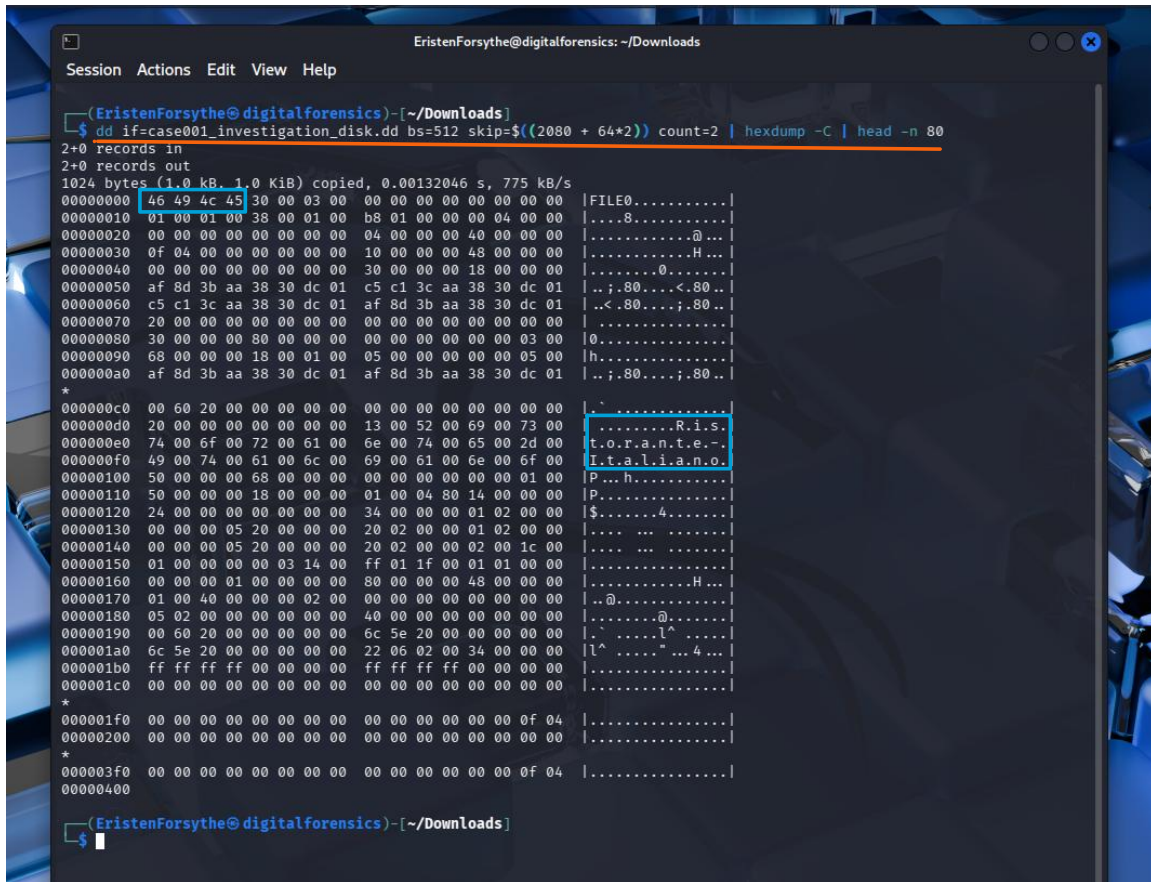
Record N sector = $2080 + (N \times 2)$

- Record 64 was extracted and manually inspected. The hexdump showed the NTFS record signature 46 49 4C 45 and the UTF-16LE filename bytes corresponding to Ristorante-Italiano.
- Record 65 was extracted and manually inspected. The hexdump showed the file signature and the UTF-16LE filename bytes corresponding to registrar.

- Ristorante-Italiano
- registrar

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```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

(EristenForsythe@digitalforensics)-[~/Downloads]
$ dd if=case001_investigation_disk.dd bs=512 skip=$((2080 + 64*2)) count=2 | hexdump -C | head -n 80
2+0 records in
2+0 records out
1024 bytes (1.0 kB, 1.0 KiB) copied, 0.00132046 s, 775 kB/s
00000000 46 49 4c 45 30 00 03 00 00 00 00 00 00 00 00 00 |FILE0.....|
00000010 01 00 01 00 38 00 01 00 b8 01 00 00 00 04 00 00 |...8.....|
00000020 00 00 00 00 00 00 00 00 04 00 00 00 40 00 00 00 |.....@...|
00000030 0f 04 00 00 00 00 00 00 10 00 00 00 48 00 00 00 |.....H...|
00000040 00 00 00 00 00 00 00 00 30 00 00 00 18 00 00 00 |.....0...|
00000050 af 8d 3b aa 38 30 dc 01 c5 c1 3c aa 38 30 dc 01 |...;.80...<.80..|
00000060 c5 c1 3c aa 38 30 dc 01 af 8d 3b aa 38 30 dc 01 |...<.80...;.80..|
00000070 20 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
00000080 30 00 00 00 80 00 00 00 00 00 00 00 00 00 03 00 |0.....|
00000090 68 00 00 00 18 00 01 00 05 00 00 00 00 00 05 00 |h.....|
000000a0 af 8d 3b aa 38 30 dc 01 af 8d 3b aa 38 30 dc 01 |...;.80...;.80..|
*
000000c0 00 60 20 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
000000d0 20 00 00 00 00 00 00 00 13 00 52 00 69 00 73 00 |.....R.i.s.|
000000e0 74 00 6f 00 72 00 61 00 6e 00 74 00 65 00 2d 00 |t.o.r.a.n.t.e.-.|
000000f0 49 00 74 00 61 00 6c 00 69 00 61 00 6e 00 6f 00 |I.t.a.l.i.a.n.o.|
00000100 50 00 00 00 68 00 00 00 00 00 00 00 00 00 01 00 |P...h.....|
00000110 50 00 00 00 18 00 00 00 01 00 04 80 14 00 00 00 |P...h.....|
00000120 24 00 00 00 00 00 00 00 34 00 00 00 01 02 00 00 |$......4.....|
00000130 00 00 00 05 20 00 00 00 20 02 00 00 01 02 00 00 |.....|
00000140 00 00 00 05 20 00 00 00 20 02 00 00 02 00 1c 00 |.....|
00000150 01 00 00 00 03 14 00 ff 01 1f 00 01 01 00 00 |.....H...|
00000160 00 00 00 01 00 00 00 00 80 00 00 00 48 00 00 00 |.....@.....|
00000170 01 00 40 00 00 00 02 00 00 00 00 00 00 00 00 00 |...@.....|
00000180 05 02 00 00 00 00 00 00 40 00 00 00 00 00 00 00 |.....@.....|
00000190 00 60 20 00 00 00 00 00 6c 5e 20 00 00 00 00 00 |l^.....l^.....4...|
000001a0 6c 5e 20 00 00 00 00 00 22 06 02 00 34 00 00 00 |.....|
000001b0 ff ff ff ff 00 00 00 00 ff ff ff ff 00 00 00 00 |.....|
000001c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
000001f0 00 00 00 00 00 00 00 00 00 00 00 00 00 0f 04 |.....|
00000200 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
000003f0 00 00 00 00 00 00 00 00 00 00 00 00 00 0f 04 |.....|
00000400
```

MFT start sector = 2080

MFT record size = 1024 bytes = 2 sectors

Record 64 sector = $2080 + (64 * 2) = 2208$

46 49 4C 45 = FILE signature = valid MFT record

Unicode filename shown: Ristorante-Italiano

Record 64 corresponds to the file “Ristorante-Italiano”.

Screenshot: record 65 hexdump showing file name for registrar.

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```
EristenForsythe@digitalforensics: ~/Downloads
(EristenForsythe@digitalforensics) - [~/Downloads]
$ dd if=case001_investigation_disk.dd bs=512 skip=$((2080 + 65*2)) count=2 | hexdump -C | head -n 80
2+0 records in
2+0 records out
1024 bytes (1.0 kB, 1.0 KiB) copied, 0.00137938 s, 742 kB/s
00000000 46 49 4c 45 30 00 03 00 00 00 00 00 00 00 00 00 |FILE0.....|
00000010 01 00 01 00 38 00 01 00 a8 01 00 00 00 04 00 00 |...8.....|
00000020 00 00 00 00 00 00 00 00 04 00 00 00 41 00 00 00 |.....A...|
00000030 31 00 00 00 00 00 00 00 10 00 00 00 48 00 00 00 |1.....H...|
00000040 00 00 00 00 00 00 00 00 30 00 00 00 18 00 00 00 |.....0...|
00000050 00 c6 3c aa 38 30 dc 01 ab e9 3c aa 38 30 dc 01 |...<.80...<.80..|
00000060 ab e9 3c aa 38 30 dc 01 00 c6 3c aa 38 30 dc 01 |...<.80...<.80..|
00000070 20 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
00000080 30 00 00 00 70 00 00 00 00 00 00 00 00 00 03 00 |0...p.....|
00000090 54 00 00 00 18 00 01 00 05 00 00 00 00 00 05 00 |T.....|
000000a0 00 c6 3c aa 38 30 dc 01 00 c6 3c aa 38 30 dc 01 |...<.80...<.80..|
*
000000c0 00 70 01 00 00 00 00 00 00 00 00 00 00 00 00 00 |p.....|
000000d0 20 00 00 00 00 00 00 00 09 00 72 00 65 00 67 00 |.....r.e.g..|
000000e0 69 00 73 00 74 00 72 00 61 00 72 00 18 00 00 00 |i.s.t.r.a.r....|
000000f0 50 00 00 00 68 00 00 00 00 00 00 00 00 00 01 00 |P...h.....|
00000100 50 00 00 00 18 00 00 00 01 00 04 80 14 00 00 00 |P.....|
00000110 24 00 00 00 00 00 00 00 34 00 00 00 01 02 00 00 |$.4.....|
00000120 00 00 00 05 20 00 00 00 20 02 00 00 01 02 00 00 |.....|
00000130 00 00 00 05 20 00 00 00 20 02 00 00 02 00 1c 00 |.....|
00000140 01 00 00 00 00 03 14 00 ff 01 1f 00 01 01 00 00 |.....|
00000150 00 00 00 01 00 00 00 00 80 00 00 00 48 00 00 00 |.....H...|
00000160 01 00 40 00 00 00 02 00 00 00 00 00 00 00 00 00 |..@.....|
00000170 16 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 |.....@.....|
00000180 00 70 01 00 00 00 00 00 9a 69 01 00 00 00 00 00 |.p.....i.....|
00000190 9a 69 01 00 00 00 00 00 21 17 00 44 00 00 00 00 |.i.....!..D....|
000001a0 ff ff ff ff 00 00 00 00 ff ff ff ff 00 00 00 00 |.....|
000001b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
000001f0 00 00 00 00 00 00 00 00 00 00 00 00 00 31 00 |.....1..|
00000200 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
000003f0 00 00 00 00 00 00 00 00 00 00 00 00 00 31 00 |.....1..|
00000400
```

Record 65 sector = $2080 + (65 * 2) = 2210$

46 49 4C 45 = FILE signature = valid MFT record

Unicode filename shown: registrar

Record 65 corresponds to the file registrar (recovered as registrar.zip).

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4.1.4 File Inventory

The following files were identified and manually recovered from the NTFS partition.

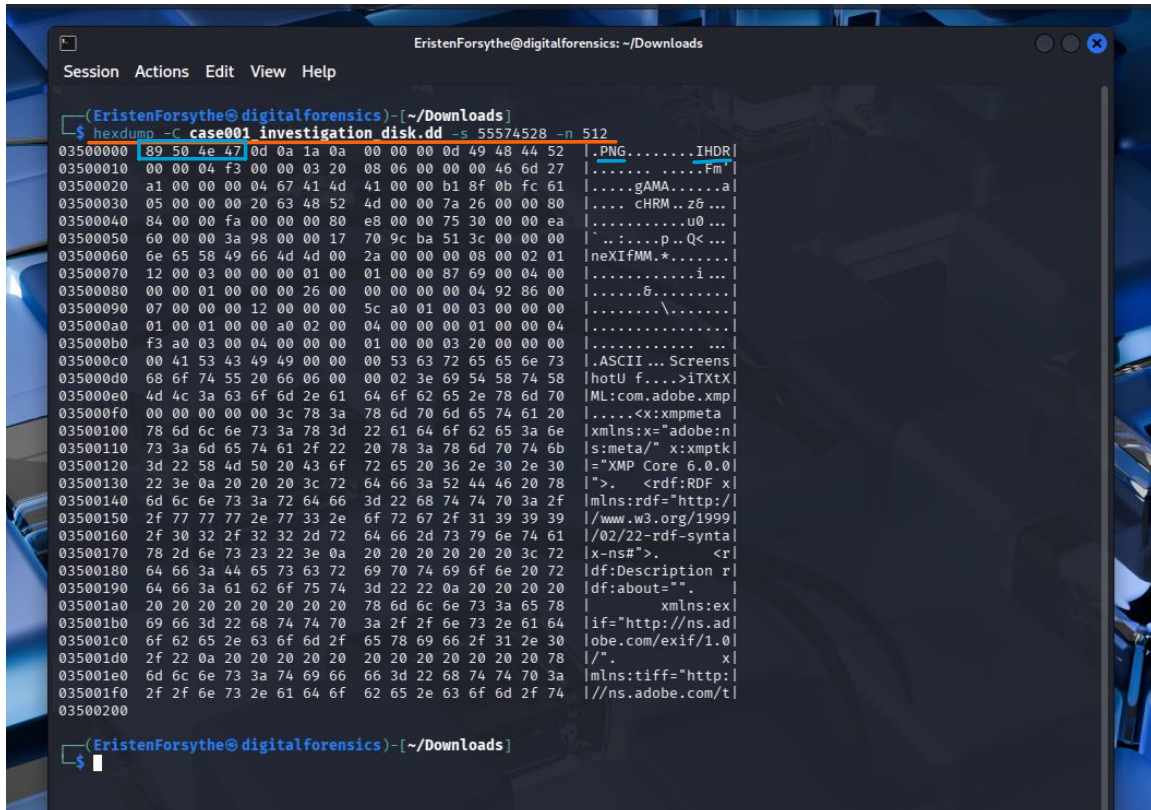
File Name	Size (Bytes)	Start offset (Bytes)	End offset (Bytes)	Status
Ristorante-Italiano	2,121,324	55574528	57695851	Active
registrar (recovered as registrar.zip)	92,570	72351744	72444313	Active

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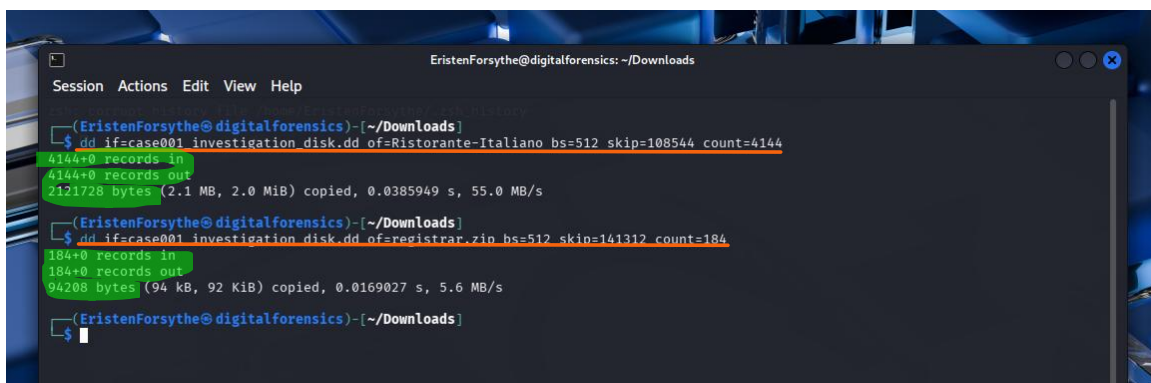
4.1.5 Manual File Recovery Process

Screenshot: Ristorante-Italiano confirmation hexdump, dd recovery, and file/hash verification.



```
(EristenForsythe@digitalforensics)~[~/Downloads]
$ hexdump -C case001_investigation_disk.dd -s 55574528 -n 512
03500000 89 50 4e 47 0d 0a 1a 0a 00 00 00 0d 49 48 44 52 |.PNG.....IHDR|
03500010 00 00 04 f3 00 00 03 20 08 06 00 00 00 46 6d 27 |.....Fm'|
03500020 a1 00 00 00 04 67 41 4d 41 00 00 b1 8f 0b fc 61 |.....gAMA....a|
03500030 05 00 00 00 20 63 48 52 4d 00 00 7a 26 00 00 80 |....cHRM..z6...|
03500040 84 00 00 fa 00 00 00 80 e8 00 00 75 30 00 00 ea |.....u0...|
03500050 60 00 00 3a 98 00 00 17 70 9c ba 51 3c 00 00 00 |'.....p..Q<...|
03500060 6e 65 58 49 66 4d 4d 00 2a 00 00 00 08 00 02 01 |neXIfMM.*.....|
03500070 12 00 03 00 00 00 01 00 01 00 00 87 69 00 04 00 |.....i...|
03500080 00 00 01 00 00 00 26 00 00 00 00 04 92 86 00 |.....G.....|
03500090 07 00 00 00 12 00 00 00 5c a0 01 00 03 00 00 00 |.....\.....|
035000a0 01 00 01 00 00 a0 02 00 04 00 00 00 01 00 00 04 |.....|
035000b0 f3 a0 03 00 04 00 00 00 01 00 00 03 20 00 00 00 |.....|
035000c0 00 41 53 43 49 49 00 00 00 53 63 72 65 65 6e 73 |.ASCII...Screens|
035000d0 68 6f 74 55 20 66 06 00 00 02 3e 69 54 58 74 58 |hotU f.....iTXt|
035000e0 4d 4c 3a 63 6f 6d 2e 61 64 6f 62 65 2e 78 6d 70 |ML:com.adobe.xmp|
035000f0 00 00 00 00 00 3c 78 3a 78 6d 70 6d 65 74 61 20 |.....<x:xmpmeta |
03500100 78 6d 6c 6e 73 3a 78 3d 22 61 64 6f 62 65 3a 6e |xmlns:x="adobe:n|
03500110 73 3a 6d 65 74 61 2f 22 20 78 3a 78 6d 70 74 6b |s:meta/" x:xmptk|
03500120 3d 22 58 4d 50 20 43 6f 72 65 20 36 2e 30 2e 30 |="XMP Core 6.0.0|
03500130 22 3e 0a 20 20 20 3c 72 64 66 3a 52 44 46 20 78 |!>. <rdf:RDF x|
03500140 6d 6c 6e 73 3a 72 64 66 3d 22 68 74 74 70 3a 2f |mlns:rdf="http:/|
03500150 2f 77 77 77 2e 77 33 2e 6f 72 67 2f 31 39 39 39 |/www.w3.org/1999|
03500160 2f 30 32 2f 32 32 2d 72 64 66 2d 73 79 6e 74 61 |/02/22-rdf-synta|
03500170 78 2d 6e 73 23 22 3e 0a 20 20 20 20 20 20 3c 72 |x-ns#">. <r|
03500180 64 66 3a 44 65 73 63 72 69 70 74 69 6f 6e 20 72 |df:Description r|
03500190 64 66 3a 61 62 6f 75 74 3d 22 22 0a 20 20 20 20 |df:about="". |
035001a0 20 20 20 20 20 20 20 20 78 6d 6c 6e 73 3a 65 78 | xmlns:ex|
035001b0 69 66 3d 22 68 74 74 70 3a 2f 2f 6e 73 2e 61 64 |if="http://ns.ad|
035001c0 6f 62 65 2e 63 6f 6d 2f 65 78 69 66 2f 31 2e 30 |lobe.com/exif/1.0|
035001d0 2f 22 0a 20 20 20 20 20 20 20 20 20 20 20 20 78 |/". x|
035001e0 6d 6c 6e 73 3a 74 69 66 66 3d 22 68 74 74 70 3a |mlns:tiff="http:|
035001f0 2f 2f 6e 73 2e 61 64 6f 62 65 2e 63 6f 6d 2f 74 |//ns.adobe.com/t|
03500200
```

89 50 4E 47 = PNG file signature



```
(EristenForsythe@digitalforensics)~[~/Downloads]
$ dd if=case001_investigation_disk.dd of=Ristorante-Italiano bs=512 skip=108544 count=4144
4144+0 records in
4144+0 records out
2121728 bytes (2.1 MB, 2.0 MiB) copied, 0.0385949 s, 55.0 MB/s

(EristenForsythe@digitalforensics)~[~/Downloads]
$ dd if=case001_investigation_disk.dd of=registrar.zip bs=512 skip=141312 count=184
184+0 records in
184+0 records out
94208 bytes (94 kB, 92 KiB) copied, 0.0169027 s, 5.6 MB/s

(EristenForsythe@digitalforensics)~[~/Downloads]
$
```

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bs=512 reads sector-sized blocks

skip = starting sector for the file

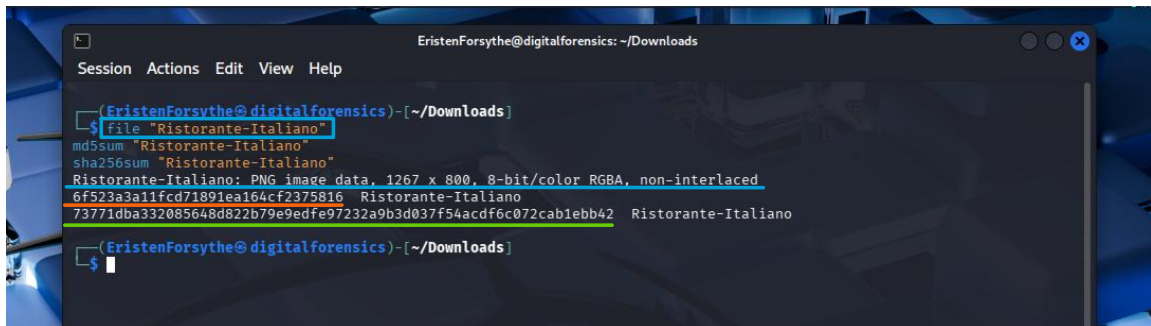
count = number of sectors extracted (allocated range

Ristorante-Italiano:

skip=108544 count=4144 = $4,144 * 512 = 2,121,728$ bytes extracted

registrar.zip:

skip=141312 count=184 = $184 * 512 = 94,208$ bytes extracted



```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help
(EristenForsythe@digitalforensics)-[~/Downloads]
$ file "Ristorante-Italiano"
md5sum "Ristorante-Italiano"
sha256sum "Ristorante-Italiano"
Ristorante-Italiano: PNG image data, 1267 x 800, 8-bit/color RGBA, non-interlaced
6f523a3a11fcd71891ea164cf2375816 Ristorante-Italiano
73771dba332085648d822b79e9edfe97232a9b3d037f54acdf6c072cab1ebb42 Ristorante-Italiano
(EristenForsythe@digitalforensics)-[~/Downloads]
$
```

file confirms true type by signature: PNG image data

md5sum and sha256sum recorded

Screenshot: registrar.zip confirmation hexdump, dd recovery, and file/hash verification.

COMP 5350/6350 Hands-On #2 Report

Student Name: Eristen Forsythe

```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

(EristenForsythe@digitalforensics)-[~/Downloads]
$ hexdump -C case001_investigation_disk.dd -s 72351744 -n 512
04500000 50 4b 03 04 14 00 00 00 08 00 17 61 3b 5b 13 2c |PK.....a;[.,|
04500010 b7 d1 27 51 01 00 cf 54 01 00 0f 00 1c 00 6e 75 |.. 'Q... T.....nu
04500020 6d 62 65 72 5f 72 6f 6f 6d 2e 6a 70 67 55 54 09 |mber_room.jpgUT.|
04500030 00 03 1e 1a d8 68 d1 35 d8 68 75 78 0b 00 01 04 |.....h.5.hux...|
04500040 e8 03 00 00 04 e8 03 00 00 9c fd 05 40 9c 41 b3 |.....@.A..|
04500050 28 88 0e ee ee ee ee ee ee ee ee ee ee 9e e0 30 |(. .....0|
04500060 b8 3b c1 1d 82 7b 08 ee ee 04 42 82 7b 70 97 37 |.;... {....B.{p.7|
04500070 fc e7 ec bd 77 e5 bd b7 bb 9d 40 8a ee ea ea fa |.....@.....|
04500080 ca 6b a6 87 7c ac 7d fc 01 a0 ca 4b cb 49 03 c0 |.k..|.}...K.I..|
04500090 c0 00 00 30 d0 1f c0 c7 0e a2 ae 9c 84 84 89 aa |... 0.....|
045000a0 ba 8a b4 9c a2 d4 7f 16 10 65 15 ed 9c 5d c0 51 |.....e... ].Q|
045000b0 01 00 27 67 4f 77 75 19 71 52 5d 3d 7d 52 98 19 |.. 'gOwu.qR]=}R..|
045000c0 00 38 00 0e 00 0d 60 03 00 cc 2c 3c 5c 95 34 a4 |.8.....'... ,<\.4.|
045000d0 35 01 a0 21 27 25 41 ea 01 42 02 fc ef c6 fd ca |5..!'%A..B.....|
045000e0 27 75 00 60 91 49 56 95 94 14 f0 ff 6c a0 59 b8 |'u..'.IV.....l.Y.|
045000f0 ba 7b 82 18 54 05 c1 1c 96 56 1e 16 20 38 1c 04 |;..T...V... 8..|
04500100 3b fa 78 ba 7e ce 5f 82 60 4c 73 87 4f 18 1c f2 |;..x~... 'ls.0...|
04500110 13 76 07 31 08 82 71 3f 61 9b ff 82 19 ff 83 f3 |.v..l..q?a.....|
04500120 5f b0 e8 27 6c e9 e4 6c 09 82 3f 79 76 b5 74 b2 |... 'l...l...?yv.t.|
04500130 fc 84 87 40 70 b4 b7 97 15 08 86 50 04 c1 91 de |...@p.....P.....|
04500140 76 56 3e 20 78 09 04 53 3a 7a 39 d9 81 e0 c7 cf |vV> x...S:z9.....|
04500150 bd 4e 56 66 1e 00 00 24 e2 e7 bc a7 95 85 2d 08 |.NVf ... $. .....-|
04500160 66 05 c1 88 ee 9a ea 12 20 58 10 00 80 45 b4 f9 |f..... X... E..|
04500170 5f 60 f3 ff 05 f6 b4 f2 f5 fc 7c 28 09 17 57 3f |.. .....|(. ..W?|
04500180 77 3b 1b 5b 4f 52 5a 0b 3a 52 36 3e 3e 5e 52 59 |w;..[ORZ.:R6>>"RY|
04500190 2b 1f 47 2b 4f 4f 26 55 33 0b 07 33 77 4b 52 09 |+.G+00G0U3..3wKR.|
045001a0 17 27 57 33 67 3f 00 e0 bf 9e f9 3f 03 fd 53 b6 |.. 'W3g?.....?..S..|
045001b0 a4 20 21 73 b3 f1 71 73 33 b1 33 b3 fd 2f 82 fa |. !s..qs3.3../..|
045001c0 ff b9 f8 7f 73 7c ea f6 bf a0 1b b5 ff e8 0c 0c |...s|.....|
045001d0 7b ea 7f ce fd 5f e1 b9 14 03 00 bc 77 20 d9 24 |{.....w..$.|
045001e0 fd cf 39 f3 6c 00 a0 3d 02 00 c0 dd f8 9f 73 94 |..9.l...=.....s.|
045001f0 df 00 00 14 90 de da 66 ff 97 e7 c1 fe b4 17 5b |.....f.....|
04500200

(EristenForsythe@digitalforensics)-[~/Downloads]
$
```

50 4B 03 04 = ZIP file signature (PK)

```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

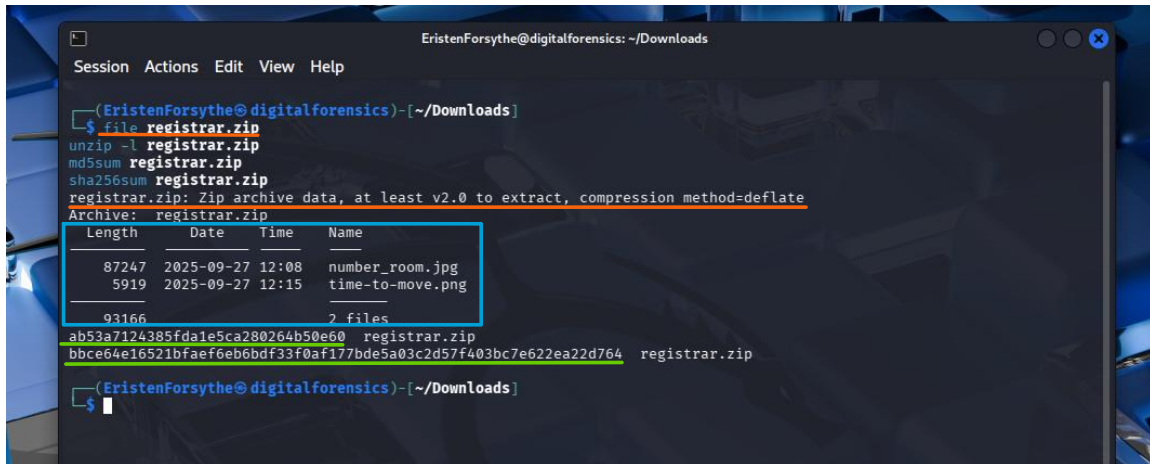
(EristenForsythe@digitalforensics)-[~/Downloads]
$ dd if=case001_investigation_disk.dd bs=512 skip=207368 count=4 of=LOCATION
4+0 records in
4+0 records out
2048 bytes (2.0 kB, 2.0 KiB) copied, 0.00205487 s, 997 kB/s

(EristenForsythe@digitalforensics)-[~/Downloads]
$ dd if=case001_investigation_disk.dd bs=512 skip=207372 count=3408 of=COA.ZIP
3408+0 records in
3408+0 records out
1744896 bytes (1.7 MB, 1.7 MiB) copied, 0.0312161 s, 55.9 MB/s

(EristenForsythe@digitalforensics)-[~/Downloads]
$
```

COMP 5350/6350 Hands-On #2 Report

Student Name: Eristen Forsythe



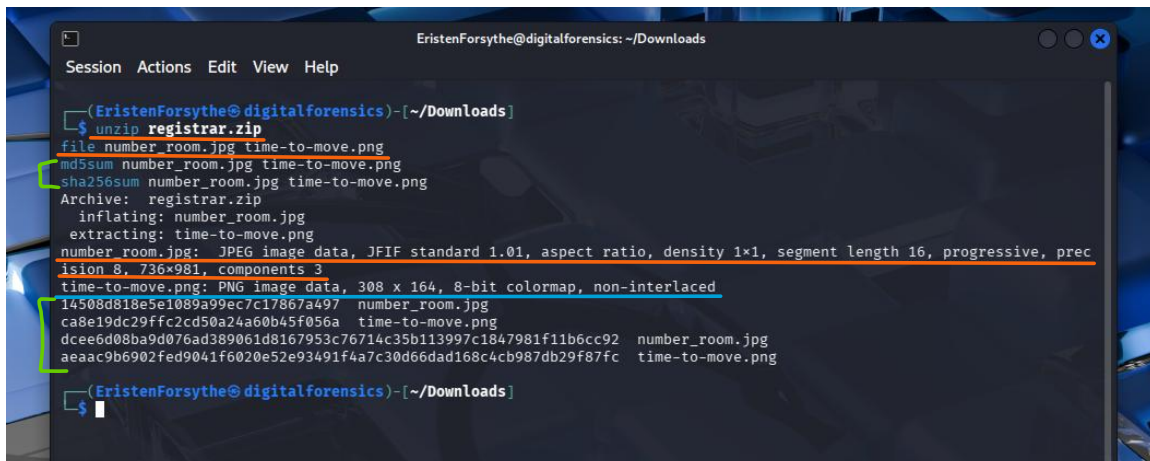
```
(EristenForsythe@digitalforensics)-[~/Downloads]
$ file registrar.zip
unzip -l registrar.zip
md5sum registrar.zip
sha256sum registrar.zip
registrar.zip: Zip archive data, at least v2.0 to extract, compression method=deflate
Archive:  registrar.zip
  Length      Date    Time    Name
  ----      -
  87247     2025-09-27 12:08    number_room.jpg
  5919     2025-09-27 12:15    time-to-move.png
  93166
  ----
  2 files
ab53a7124385fdale5ca280264b50e60 registrar.zip
bbce64e16521bfaef6eb6bdf33f0af177bde5a03c2d57f403bc7e622ea22d764 registrar.zip
(EristenForsythe@digitalforensics)-[~/Downloads]
```

file confirms ZIP by signature

unzip -l lists embedded artifacts: number_room.jpg and time-to-move.png

md5sum and sha256sum recorded

Screenshot: Extraction of registrar.zip and file/hash verification of number_room.jpg and time-to-move.png.



```
(EristenForsythe@digitalforensics)-[~/Downloads]
$ unzip registrar.zip
file number_room.jpg time-to-move.png
md5sum number_room.jpg time-to-move.png
sha256sum number_room.jpg time-to-move.png
Archive:  registrar.zip
  inflating: number_room.jpg
  extracting: time-to-move.png
number_room.jpg:  JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length 16, progressive, precision 8, 736x981, components 3
time-to-move.png: PNG image data, 308 x 164, 8-bit colormap, non-interlaced
14508d818e5e1089a99ec7c17867a497 number_room.jpg
ca8e19dc29ffc2cd50a24a60b45f056a time-to-move.png
dcee6d08ba9d076ad389061d8167953c76714c35b113997c1847981f11b6cc92 number_room.jpg
aeaac9b6902fed9041f6020e52e93491f4a7c30d66dad168c4cb987db29f87fc time-to-move.png
(EristenForsythe@digitalforensics)-[~/Downloads]
```

unzip registrar.zip gets the embedded artifacts.

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Student Name: Eristen Forsythe

number_room.jpg:

file confirms JPEG image type by signature

md5sum and sha256sum recorded

time-to-move.png:

file confirms PNG image type by signature

md5sum and sha256sum recorded

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Student Name: Eristen Forsythe

4.1.6 Recovered Files Summary

Ristorante-Italiano – recovered and viewable

Registrar.zip – recovered and viewable

Recovered files are included in the submission directory

COMP 5350/6350 Hands-On #2 Report

Student Name: Eristen Forsythe

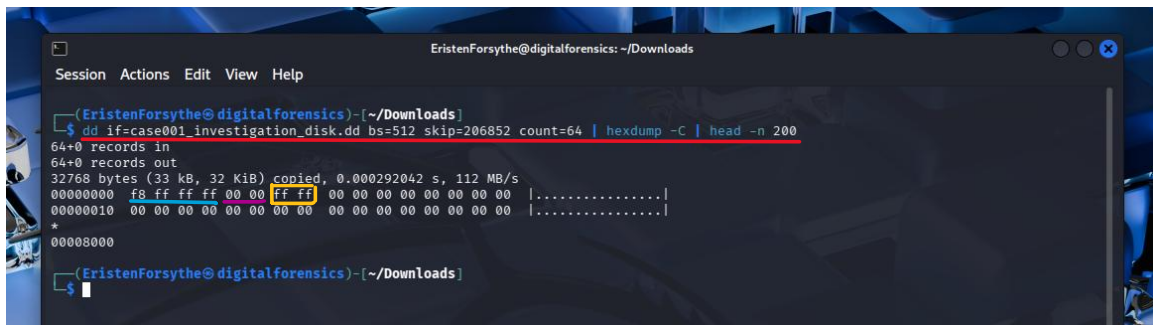
4.2 Partition 2 Analysis (FAT16)

4.2.1 Partition Description

- File system type: FAT16
- Starting and ending offsets: Sector 206848 (105906176 bytes) & Sector 452607 (231735295 bytes)
- Intended purpose (if inferable): additional User storage

4.2.2 FAT Structure Analysis

This FAT16 partition uses 512-byte sectors and 4 sectors per cluster (cluster size 2,048 bytes). The FAT16 shows a mostly contiguous pattern for the small active LOCATION file and a lot bigger allocation matching to the COA.ZIP. Even though COA.ZIP was deleted (E5 marker in the root directory), its data clusters were not overwritten, so the file remained recoverable from its referenced data area using dd.



```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

(EristenForsythe@digitalforensics)-[~/Downloads]
$ dd if=case001_investigation_disk.dd bs=512 skip=206852 count=64 | hexdump -C | head -n 200
64+0 records in
64+0 records out
32768 bytes (33 kB, 32 KiB) copied, 0.000292042 s, 112 MB/s
00000000 f8 ff ff ff 00 00 ff ff 00 00 00 00 00 00 00 00 |.....|
00000010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
00000800
(EristenForsythe@digitalforensics)-[~/Downloads]
$
```

FAT #1 Evidence (FAT16):

f8 ff ff ff = reserved/media descriptor region

Free/ Unallocated clusters = 00 00

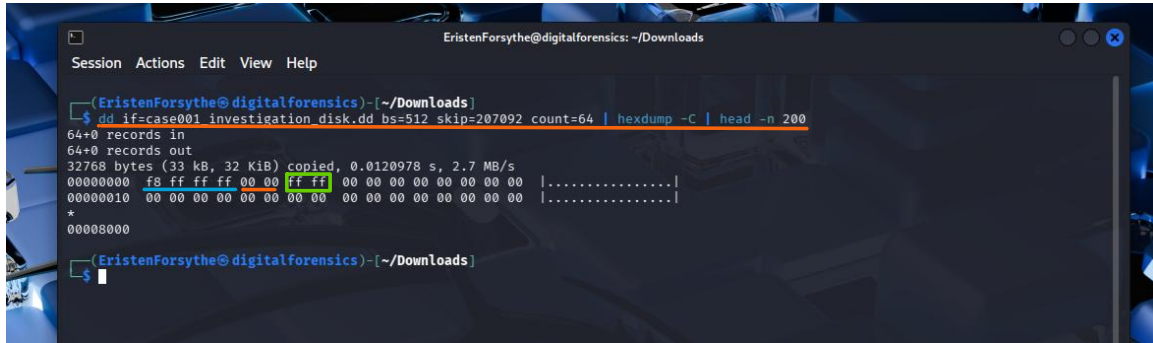
EOC / end-of-chain marker observed: ff ff

Cluster size = $4 * 512 = 2,048$ bytes

Data area start sector = 207,364

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```
Session Actions Edit View Help
(EristenForsythe@digitalforensics)-[~/Downloads]
$ dd if=case001_investigation_disk.dd bs=512 skip=207092 count=64 | hexdump -C | head -n 200
64+0 records in
64+0 records out
32768 bytes (33 kB, 32 KiB) copied, 0.0120978 s, 2.7 MB/s
00000000 f8 ff ff ff 00 00 ff ff 00 00 00 00 00 00 00 00 |.....|
00000010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
00000800
(EristenForsythe@digitalforensics)-[~/Downloads]
$
```

FAT #2 Evidence (FAT16):

f8 ff ff ff = reserved/media descriptor region

Free/ Unallocated clusters = 00 00

EOC / end-of-chain marker observed: ff ff

FAT #2 matches FAT #1 structure, confirming the duplicate FAT copy exists and supports filesystem integrity.

Region	Start Sector	End Sector	Size (sectors)	Start Offset (bytes)
Reserved Area (Boot + Reserved)	206,848	206,851	4	105,906,176
FAT #1	206,852	207,091	240	105,908,224
FAT #2	207,092	207,331	240	106,031,104
Root Directory	207,332	207,363	32	106,153,984
Data Area	207,364	452,607	245,244	106,170,368

(End Sector = Start Sector + Size - 1. Start Offset = Start Sector × 512.)

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4.2.3 Root Directory Analysis

The FAT16 root directory contained two entries: LOCATION (active) and COA.ZIP (deleted). The deleted COA.ZIP entry retained the starting cluster and file size, confirming the file was recoverable with manual dd extraction even after deletion.

```
EristenForsythe@digitalforensics: ~/Downloads
$ dd if=case001_investigation_disk.dd bs=512 skip=207332 count=32 | hexdump -C | head -n 200
32+0 records in
32+0 records out
00000000 41 6c 00 6f 00 63 00 61 00 74 00 0f 00 08 69 00 |A.l.o.c.a.t...i.|
00000010 6f 00 6e 00 00 00 ff ff ff ff 00 00 ff ff ff ff |o.n.....+|
00000020 4c 4f 43 41 54 49 4f 4e 20 20 20 20 00 7e d2 2b |LOCATION.....+|
00000030 3c 5b 3c 5b 00 00 d2 2b 3c 5b 03 00 62 00 00 00 |<[<[...+<[.b...|
00000040 e5 43 00 6f 00 41 00 7e 00 7a 00 0f 00 6c 69 00 |.C.o.A...z...li.|
00000050 70 00 00 00 ff ff ff ff ff ff 00 00 ff ff ff ff |p.....+|
00000060 e5 4f 41 20 20 20 20 20 5a 49 50 20 00 7e d2 2b |.OA...ZIP...+|
00000070 3c 5b 3c 5b 00 00 d2 2b 3c 5b 04 00 1e 9f 1a 00 |<[<[...+<[.....|
00000080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
16384 bytes (16 kB, 16 KiB) copied, 0.00547116 s, 3.0 MB/s*
00004000
EristenForsythe@digitalforensics: ~/Downloads
$
```

Directory entries present (from root directory hexdump):

LOCATION is active (normal first filename byte).

COA.ZIP is deleted because the first filename byte is E5 (deleted marker).

LOCATION:

03 00 = 0x0003 = starting cluster

62 00 00 00 (little-endian) = 0x00000062 = 98 bytes = file size

Created on 2025-09-28 05:30:36 (time bytes d2 2b, date bytes 3c 5b)

Last modified on 2025-09-28 05:30:36 (time bytes d2 2b, date bytes 3c 5b)

Last accessed on 2025-09-28 (date bytes 3c 5b)

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Student Name: Eristen Forsythe

COA.ZIP (deleted):

E5 = deleted marker

04 00 = 0x0004 = starting cluster

1E 9F 1A 00 (little-endian) = 0x001A9F1E = 1,744,670 bytes = real file size

Created on 2025-09-28 05:30:36 (time bytes d2 2b, date bytes 3c 5b)

Last modified on 2025-09-28 05:30:36 (time bytes d2 2b, date bytes 3c 5b)

Last accessed on 2025-09-28 (date bytes 3c 5b)

- COA.ZIP is a deleted entry (E5 marker), but the starting cluster (04 00) and file size (1E 9F 1A 00) are still present, so manual recovery is possible.
-

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Student Name: Eristen Forsythe

4.2.4 File Inventory

File Name	Size (Bytes)	Start Offset (Bytes)	End Offset (Bytes)	Status
LOCATION	98	106172416	106172513	Active
COA.ZIP	1744670	106174464	107919133	Deleted

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4.2.5 Manual File Recovery Process

Files were recovered by calculating sector and byte offsets derived from FAT16 boot sector values and root directory analysis. LOCATION was extracted using its starting cluster and recovered as ASCII text (using Base64). COA.ZIP was extracted from its starting cluster as an allocated range and then opened using the password found from other evidence (restaurant ZIP code = 75018).

```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help
(EristenForsythe@digitalforensics)~/Downloads
$ dd if=case001_investigation_disk.dd bs=512 skip=207368 count=4 of=LOCATION
4+0 records in
4+0 records out
2048 bytes (2.0 kB, 2.0 KiB) copied, 0.00205487 s, 997 kB/s
(EristenForsythe@digitalforensics)~/Downloads
$ dd if=case001_investigation_disk.dd bs=512 skip=207372 count=3408 of=COA.ZIP
3408+0 records in
3408+0 records out
1744896 bytes (1.7 MB, 1.7 MiB) copied, 0.0312161 s, 55.9 MB/s
(EristenForsythe@digitalforensics)~/Downloads
$
```

```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help
(EristenForsythe@digitalforensics)~/Downloads
$ unzip COA.zip
unzip: cannot find or open COA.zip, COA.zip.zip or COA.zip.ZIP.
(EristenForsythe@digitalforensics)~/Downloads
$ unzip COA.ZIP
Archive: COA.ZIP
[COA.ZIP] sunset_on_ice.png password:
replace sunset_on_ice.png? [y]es, [n]o, [A]ll, [N]one, [r]ename: n
(EristenForsythe@digitalforensics)~/Downloads
$
```

```
(EristenForsythe@digitalforensics)~/Downloads
$ md5sum LOCATION
sha256sum LOCATION
md5sum COA.ZIP
sha256sum COA.ZIP
6ff34443f8736686fa73767532903a97 LOCATION
b377bba13e79f85c6cc78555af4eaf019e406544ef5735ff55b31c50f995bd40 LOCATION
aaeca3a57133d582d4f207c032fe5279 COA.ZIP
1a41b7c404cf8df70a5f13df9513fe476f55b5642e2a09386d95aaa30ecbe907 COA.ZIP
(EristenForsythe@digitalforensics)~/Downloads
$
```

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Student Name: Eristen Forsythe

Boot Sector Values:

Partition Start = 206,848

Bytes/Sector = 512

Sectors/Cluster = 4

Reserved Sectors = 4

FATs = 2

Sectors/FAT = 240

Root Directory = 32 sectors

Cluster Size = $4 \times 512 = 2,048$ bytes

Data Area Start: $206,848 + 4 + (2 \times 240) + 32 + 4 = 206,848 + 4 + 480 + 32 + 4 = 207,364$

LOCATION:

Cluster Start = 3

Start Sector = $207,364 + ((3-2) \times 4) = 207,364 + 4 = 207,368$

Start Byte = $207,368 \times 512 = 106,172,416$

File Size = 98 bytes

Allocated Clusters = $\text{ceil}(98 / 2,048) = 1$

Allocated Sectors = $1 \times 4 = 4$

Allocated Bytes = $4 \times 512 = 2,048$

End Byte (true end) = $106,172,416 + 98 - 1 = 106,172,513$

Allocated End Byte = $106,172,416 + 2,048 - 1 = 106,174,463$

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COA.ZIP (deleted):

Cluster Start = 4

Start Sector = $207,364 + ((4-2) \times 4) = 207,364 + 8 = 207,372$

Start Byte = $207,372 \times 512 = 106,174,464$

Real File Size = 1,744,670 bytes (from directory entry)

Allocated Sectors extracted = 3,416 (to cover allocated range)

Allocated Bytes = $3,416 \times 512 = 1,748,992$

End Byte (true end) = $106,174,464 + 1,744,670 - 1 = 107,919,133$

Allocated End Byte = $106,174,464 + 1,748,992 - 1 = 107,923,455$

Manual recovery commands (dd):

```
dd if=case001_investigation_disk.dd bs=512 skip=207368 count=4 of=LOCATION
```

```
dd if=case001_investigation_disk.dd bs=512 skip=207372 count=3408 of=COA.ZIP
```

Confirm successful recovery:

LOCATION recovered as ASCII text and successfully decoded using:

```
base64 -d LOCATION
```

COA.ZIP recovered as a ZIP archive (even though deleted) and successfully extracted using: `unzip -P 75018 COA.ZIP -d coa_unzipped/ file coa_unzipped/*`

COA.ZIP real size from directory entry = 1,744,670 bytes, but extraction used the allocated sector range (1,748,992 bytes) to ensure full recovery.

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Student Name: Eristen Forsythe

4.2.6 Recovered Files Summary

LOCATION – recovered and extracted

COA.ZIP - recovered and extracted

Recovered files are included in the submission directory

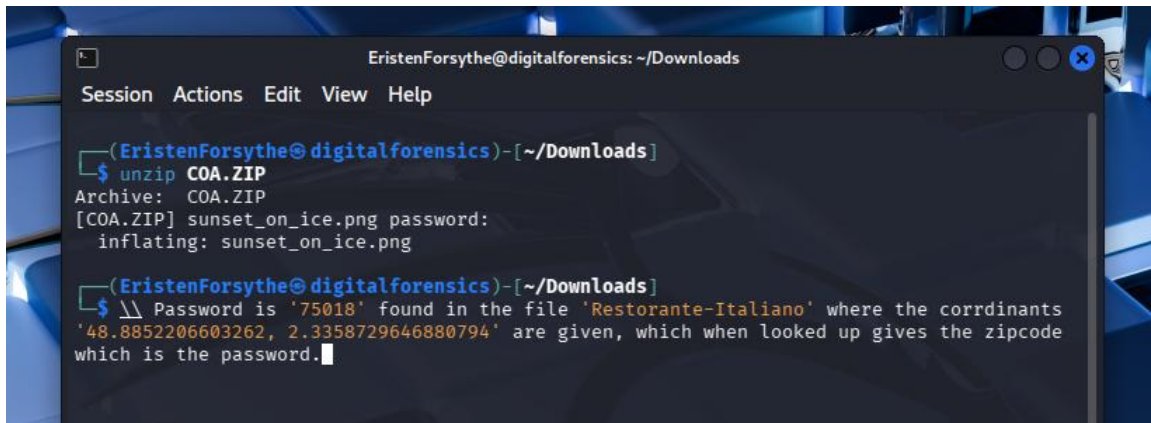
COMP 5350/6350 Hands-On #2 Report

Student Name: Eristen Forsythe

5. Cross-Partition Evidence Analysis

Evidence from both partitions forms a single workflow used by the suspects. The FAT16 partition has LOCATION (base64-encoded text) and the encrypted COA.ZIP archive. LOCATION provided the Paris hotel location (“Le Burgundy Paris”) and coordinates (48.8852206603262, 2.3358729646880794). The NTFS partition has registrar.zip, which contained two images indicating the room number (332) and the time to act (12:40 PM). The password requirement in the intercepted message linked the restaurant zipcode to the COA zip. The correct password to the zip is 75018, allowing extraction of the Certificate of Authenticity document.

Screenshot: unzip of COA.ZIP showing successful extraction of sunset_on_ice.png.



```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

(EristenForsythe@digitalforensics)-[~/Downloads]
$ unzip COA.ZIP
Archive: COA.ZIP
[COA.ZIP] sunset_on_ice.png password:
inflating: sunset_on_ice.png

(EristenForsythe@digitalforensics)-[~/Downloads]
$ \ Password is '75018' found in the file 'Restorante-Italiano' where the corrdinants
'48.8852206603262, 2.3358729646880794' are given, which when looked up gives the zipcode
which is the password.
```

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6. Content Analysis of Recovered Files

LOCATION (base64): Decoded content revealed “Le Burgundy Paris” and GPS coordinates (48.871550902843154, 2.328319281887161).

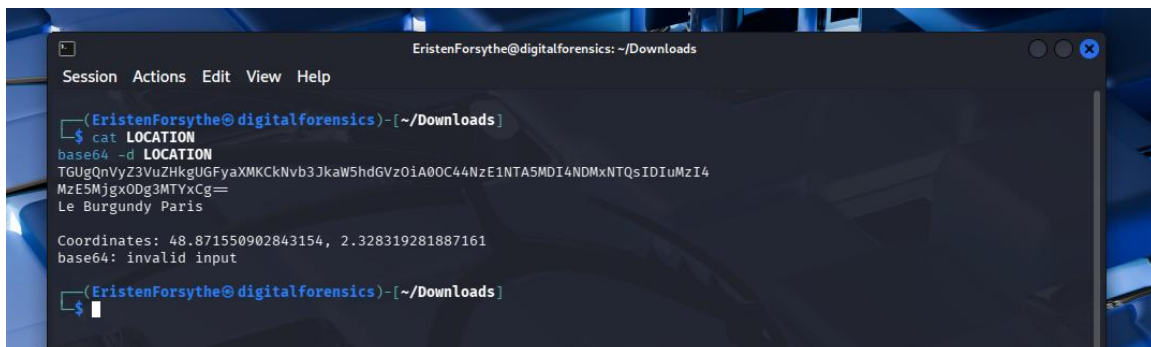
registrar.zip: Archive contained number_room.jpg (hotel door showing room 332) and time-to-move.png (display showing 12:40 PM).

COA.ZIP / sunset_on_ice.pdf: After decrypting COA.ZIP with password 75018, the file sunset_on_ice.png was extracted. The file command identified it as a PDF document; it was renamed to sunset_on_ice.pdf. The PDF is a Certificate of Authenticity for the artwork titled “Sunset on Ice,” attributed to artist Bernardette Legrand, year 2016.

Screenshot: LOCATION base64 decode output (Le Burgundy Paris + coordinates).

All recovered files

Artifact (recovered name)	file result (signature)	Correct extension
Ristorante-Italiano	PNG image data	.png
registrar.zip	Zip archive data	.zip
LOCATION	ASCII text	.txt (or no extension, its text)
COA.ZIP	Zip archive data	.zip
number_room.jpg	JPEG image data	.jpg
time-to-move.png	PNG image data	.png
sunset_on_ice.pdf	PDF document	.pdf



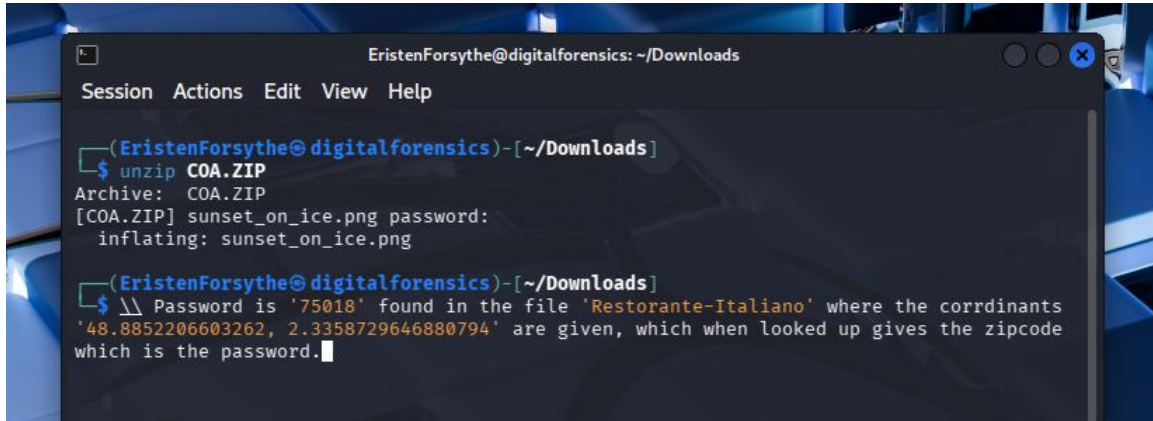
```
EristenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help
(EristenForsythe@digitalforensics) - [~/Downloads]
$ cat LOCATION
base64 -d LOCATION
TGUGQnVyZ3VuZGkgUGFyaXMKCKNvb3JkaW5hdGVzOIA0OC44NzE1NTA5MDI4NDMxNTQsIDIuMzI4
MzESMjg3ODg3MTYxCg==
Le Burgundy Paris

Coordinates: 48.871550902843154, 2.328319281887161
base64: invalid input
(EristenForsythe@digitalforensics) - [~/Downloads]
$
```

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Student Name: Eristen Forsythe

Screenshot: unzip COA.ZIP output (sunset_on_ice extracted).



```
ErystenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

(EristenForsythe@digitalforensics)-[~/Downloads]
$ unzip COA.ZIP
Archive: COA.ZIP
[COA.ZIP] sunset_on_ice.png password:
inflating: sunset_on_ice.png

(EristenForsythe@digitalforensics)-[~/Downloads]
$ \ Password is '75018' found in the file 'Restorante-Italiano' where the corrdinants
'48.8852206603262, 2.3358729646880794' are given, which when looked up gives the zipcode
which is the password.
```

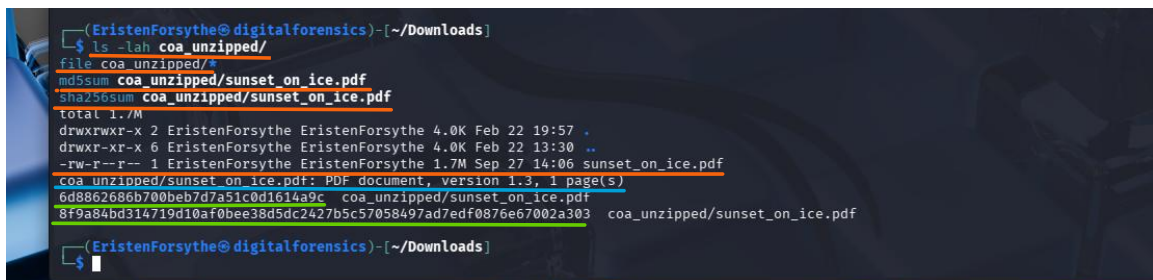
Screenshot: file identifies extracted CoA as PDF (extension spoofing).



```
ErystenForsythe@digitalforensics: ~/Downloads
Session Actions Edit View Help

(EristenForsythe@digitalforensics)-[~/Downloads]
$ file sunset_on_ice.png
mv sunset_on_ice.png sunset_on_ice.pdf
file sunset_on_ice.pdf
sunset_on_ice.png: PDF document, version 1.3, 1 page(s)
sunset_on_ice.pdf: PDF document, version 1.3, 1 page(s)

(EristenForsythe@digitalforensics)-[~/Downloads]
$
```



```
(EristenForsythe@digitalforensics)-[~/Downloads]
$ ls -lah coa_unzipped/
file coa_unzipped/+
md5sum coa_unzipped/sunset_on_ice.pdf
sha256sum coa_unzipped/sunset_on_ice.pdf
total 1.7M
drwxrwxr-x 2 EristenForsythe EristenForsythe 4.0K Feb 22 19:57 .
drwxr-xr-x 6 EristenForsythe EristenForsythe 4.0K Feb 22 13:30 ..
-rw-r--r-- 1 EristenForsythe EristenForsythe 1.7M Sep 27 14:06 sunset_on_ice.pdf
coa_unzipped/sunset_on_ice.pdf: PDF document, version 1.3, 1 page(s)
6d8862686b700beb7d7a51c0d1614a9c coa_unzipped/sunset_on_ice.pdf
8f9a84bd314719d10af0bee38d5dc2427b5c57058497ad7edf0876e67002a303 coa_unzipped/sunset_on_ice.pdf

(EristenForsythe@digitalforensics)-[~/Downloads]
$
```

ls -lah confirms the recovered file exists: sunset_on_ice.pdf

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file coa_unzipped/* confirms true signature/type: PDF document (version 1.3)

md5sum = 6d8862686b700beb7d7a51c0d1614a9c

sha256sum =

8f9a84bd314719d10af0bee38d5dc2427b5c57058497ad7edf0876e67002a303

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7. Data Hiding and Anti-Forensics Techniques

- Deleted-file misuse: COA.ZIP was deleted in the FAT16 root directory (E5 marker) but the data was not overwritten, allowing manual recovery.
 - Password protection: COA.ZIP contents were encrypted and required the restaurant ZIP code (75018) to extract.
 - Data encoding: LOCATION stored the meeting/hotel information as Base64-encoded text.
 - Nested container: registrar data was stored inside a ZIP archive (registrar.zip), requiring extraction to access the internal artifacts.
 - File renaming / extension spoofing: sunset_on_ice.png was misleading, file signature identified it as a PDF and it required renaming for correct viewing.
 - No evidence of slack space usage, steganography, or hidden partitions was found.
-

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Student Name: Eristen Forsythe

8. Timeline Reconstruction

- The criminals' goal was to get a photographic copy of the Certificate of Authenticity (CoA) for the stolen artwork so they could convince the buyers it was real.
- The intercepted message from "Shadow" shows the rules of the operation:
 - o no calls, no texts, files only, hide evidence by deleting/renaming extensions, and password-protect the final CoA file using the ZIP code of the restaurant where they met.

The files show them planning the next steps.

- LOCATION (FAT16) contained the hotel information, but it was stored as Base64 text.
- After decoding it, it revealed the hotel location "Le Burgundy Paris" and coordinates, matching Shadow's statement that they were tracking the registrar and would post hotel/room info.
- registrar.zip (NTFS) contained the move instructions as files inside the ZIP.
- Inside it, number_room.jpg showed the room number (332) and time-to-move.png showed the move time (12:40 PM).
- This gives the exact window for when they planned to enter and get the photo of the CoA.

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- COA.ZIP (FAT16) was the final target file.
- It was deleted (E5 marker in the root directory), but the starting cluster and file size were still present, so it could be manually recovered with dd.
- The archive was password protected like Shadow instructed.
- The password was 75018 (the restaurant ZIP code mentioned by Shadow's rule), and using that password successfully extracted the contents of COA.ZIP.
- After extraction, the COA file was disguised as sunset_on_ice.png, but the file signature identified it as a PDF document.
- That confirms extension spoofing was used to avoid detection while storing the CoA.

Overall, the evidence shows: they met at a restaurant (ZIP used as password), moved to Le Burgundy Paris, targeted room 332 at 12:40 PM, recovered the COA photo, encrypted it, deleted it, and disguised the final document to hide what it really was.

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9. Vulnerability Analysis

- Vulnerability description: No software exploitation artifacts, malware, or vulnerability indicators (e.g., exploit tools, exploit scripts, or CVE references) were identified in the recovered files. The operation appears to rely on physical access and operational tradecraft (file-based communications, encoding, encryption, deletion, and extension spoofing) rather than exploitation of a software flaw.
 - CVE identifier (if applicable): N/A
 - CVSS base score: N/A
 - Why this vulnerability mattered to the plan: No software vulnerability was required for the observed plan. The suspects' success depended on tracking the registrar, obtaining physical access at the hotel room, and hiding evidence through anti-forensics techniques.
-

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10. Conclusion

Manual analysis of the disk image identified and recovered key artifacts that reconstruct the suspects' plan. The evidence shows the suspects tracked the registrar's location to a Paris hotel, coordinated a move to a specific room at a specific time, and secured a photographic Certificate of Authenticity in an encrypted archive protected by the restaurant ZIP code (75018). The group attempted to reduce detection by deleting files, encoding text, nesting data in archives, encrypting the CoA, and disguising the CoA document with a misleading extension. All recovered files, hashes, and recovery offsets are provided in the accompanying RecoverySheet.xlsx and recovered file archive